MIL-F-80133B
30 October 1987
SUPERSEDING
MIL-F-80133A
28 March 1973

MILITARY SPECIFICATION

FURNACES, HEAT TREATING, PIT TYPE, ELECTRIC, FORCED CONVECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers vertical pit type, heat treating furnaces of the electrically heated, forced convection design.

1.2 Classification. The furnaces are of the following classes. The class supplied shall be as specified (see 6.2.1).

Class 1 - 350 - 1250 Degrees F Operating Temperature.

Class 2 - 350 - 1400 Degrees F Operating Temperature.

Class 3 - 600 - 1800 Degrees F Operating Temperature.

2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MIL-F-3296 - Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking), Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, 1910 - Occupational Safety and Health Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402-0001).

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA ICS 1 - Industrial Controls and Systems.

ANSI/NEMA MG 1 - Motors and Generators

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363).

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INSTRUMENT SOCIETY OF AMERICA (ISA)

MC96.1 - Temperature Measurement Thermocouples.

(Application for copies should be addressed to Instrument Society of America, P. O. Box 12277, Search Triangle Park, Durham, SC 27709-2277.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook of Industrial Loss Prevention

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston-Providence Turnpike, Norwood, MA 02062-5082).

(Technical society and technical association specification and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187).

(Non-Government standards and other publications are normally available from the organizations that prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. Furnaces and equipment provided in accordance with this specification shall be new and of the manufacturer's current design. The furnaces shall be of the vertical pit, forced convection type with a cylindrical heating chamber. The opening shall be located in the top of the furnace for inserting and removing the charge in a vertical plane. The furnace shall be electrically heated with heating elements located in the work chamber. The furnace shall be provided with either a radiation baffle or a removable solid

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side wall work container to separate the heating elements from the effective work area in accordance with 3.7.4. The furnace shall be provided with a fan to circulate the heated gas through the charge space. Unless otherwise specified the fan shall be located either directly beneath the work chamber or in a separate chamber directly connected to the work chamber. When specified (see 6.2.1), the fan may be located in the cover or directly beneath the work chamber or in a separate chamber directly connected to the work chamber.

3.2.1 Controlled atmosphere. When required, the furnace shall be designed and equipped for operation with a manufactured atmosphere. The type of atmosphere shall be as specified (see 6.2.1), and may include such types as endothermic, argon, natural gas, steam, and ammonia gas. When steam atmosphere is specified, the furnace heating chamber shall be provided with an alloy liner located between the heating elements and insulation. The liner shall provide a steam-tight seal between the chamber and insulation. Furnaces designed to operate with ammonia gas shall be provided with means for purging the ammonia out of the chamber before unloading.

3.2.2 Measurement system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the furnace. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.3 Reclaimed materials. The furnace may contain reclaimed materials provided such materials will not jeopardize the furnace's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials that normally would have been selected for use on the furnace.

3.2.4 Energy efficiency. The furnace and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the furnace that present safety hazards. The safety devices shall not interfere with the operation of the furnace. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the furnace, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 that are applicable to the furnace itself. In addition to other safety requirements specified herein, the safety system of the furnace shall conform to the requirements of FM "Handbook of Industrial Loss Prevention."

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3.2.6 Mercury restriction. The furnace shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the furnace.

3.2.8 Environmental protection. The furnace shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the furnace shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The furnace shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the furnace to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solders or pastes shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent loosing. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.3.5 Painting. Unless otherwise specified (see 6.2.1), the furnace shall be painted with heat resisting paint in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the furnace and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4 Electrical equipment. The furnace shall be furnished with electrical motors, contactors, controls, relays, transformers, signal lights and switches, pre-wired where possible for operation upon connection to the source of power.

3.4.1 Power supply. The furnace shall operate from a single source primary power supply. A fusible disconnect switch, or circuit breaker conforming to NEMA ICS 1 requirements, shall be provided to disconnect all power to the furnace. A stepdown transformer shall be provided for the 120 volt control system power supply. The furnace shall be designed and wired for operation on the voltage and phase specified (see 6.2.1).

3.4.2 Motors. Motors rated in excess of 1/4 HP shall be polyphase 230/460 volt type. Where the furnace power supply is 208 volts, 3 phase, 60 Hz, the motor shall be wound specifically for 208 voltage. Fractional horsepower motors of 1/4 HP and less shall operate on 115 volts, 60 Hz power supply. Motors shall conform to ANSI/NEMA Standard MG 1.

3.4.3 Electrical connections. Electrical connections of conductors and terminal parts shall be of the pressure of solder type. When soldered connections are used, the conductor and terminal parts shall be mechanically secured before soldering. Flux for soldering shall be of a resin base.

3.4.4 Solid-state components. Solid-state design shall be used throughout for electronic components.

3.5 Size. The size of the furnace work chamber shall be as specified (see 6.2.1), and should be selected from the dimensions listed in table I. The dimensions specified shall define the minimum inside effective working volume of the heating chamber in which performance requirements and temperature uniformity specified herein shall be maintained.

3.6 Performance. Each furnace shall be capable of continuous operation at any temperature within its operating range with maximum loads as specified in table II.

3.6.1 Recovery rate. The recovery rate shall be not less than that specified in table II. The recovery rate is based on gross load, and includes

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the weight of baskets, trays, or fixtures required to retain the workload. The recovery rate specified for the class 1 furnace is applicable only for furnaces designed for ammonia gas for use in nitriding processes. Class 1 furnaces, designed for other than nitriding processes, shall have approximately the same recovery rate as specified for the class 2 furnace. The recovery rate is determined by loading the specified load in the furnace which has been stabilized at the temperature shown in table II. If a size different from the sizes listed herein is required, the recovery rate and maximum load shall be as specified (see 6.2.1).

3.6.2 Temperature uniformity. With the furnace charged and the temperature stabilized, the temperature variation from chamber control point to any other point within the chamber working area shall not vary more than ±10 degrees Fahrenheit for class 1, or ±15 degrees Fahrenheit for class 2, or ±20 degrees Fahrenheit for class 3 furnaces. Each furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature within the operating temperature range specified for the class furnace.

3.7 Components.

3.7.1 Shell. The outer shell of the furnace shall be cylindrical in shape, fabricated from sheet steel of at least 3/32 inch in thickness and properly reinforced with steel members and shapes. Where joining by welding is required, only electric welding with continuous welds shall be used. Furnaces designed for using a manufactured atmosphere, without a retort, shall be suitably sealed by continuous welds, bolting, and gasketing around openings, to prevent air infiltration. The top and bottom plates shall be fabricated from steel plate at least 3/8 inch thickness.

3.7.2 Cover. The furnace shall be provided with an insulated cover. The insulation may be ceramic, refractory, block or bulk type insulation. The quality and thickness of the insulation shall be such that the surface temperature of the cover will be compatible with the outside surface temperature of the shell. The cover for air atmosphere furnaces shall rest on a ceramic composition gasket. The gasket shall be retained in a channel in the shell, or insulation. Furnaces designed to use a manufactured atmosphere shall be provided with a sealing means between the cover and chamber to minimize air infiltration. The cover for furnaces designed to operate with ammonia gas shall rest on a ceramic composition gasket, and the shell shall be provided with an oil seal trough around the outer perimeter. Furnaces designed to use a carburizing atmosphere shall be provided with an ore seal trough. The cover shall be provided with a mechanical mechanism for easy lifting, and shall swing horizontally in either direction from the closed position. The maximum effort required to lift, swing, or close the cover shall not exceed a force of 40 pounds. When specified (see 6.2.1), the cover shall be provided with an air activated lifting device and provide for manually swinging open to either side.

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3.7.3 Lining. The lining of the furnace heating chamber shall consist of at least 2000 degrees Fahrenheit insulating fire brick, backed with insulating slab or preformed insulation. Overall thickness and quality of the lining shall be such that the outside surface temperature of the shell and cover for class 1 and class 2 furnaces shall not exceed 100 degrees Fahrenheit above ambient, or the class 3 furnace 150 degrees Fahrenheit above ambient, after the furnace temperature has stabilized at maximum operating temperature. Except for the 12 inch diameter, class 1 furnace, powdered, shredded, or granular material shall not be used for insulating purposes. Ambient temperature is defined as 77 degrees Fahrenheit ±18 degrees Fahrenheit.

3.7.4 Baffle and work container. The furnace shall be provided with either a radiation baffle, or a removable solid side wall work container as specified (see 6.2.1).

3.7.4.1 Baffle. The radiation baffle shall be located between the heating elements and working area to prevent direct radiant heat of a source hotter than the desired work from striking the charge. The baffle shall be cast, or fabricated from a heat resisting alloy that will withstand continuous maximum temperature without sagging or warping.

3.7.4.2 Work container. The work container shall serve as the radiation baffle. The largest container or work basket for a particular furnace shall have inside dimensions equivalent to the size specified for the furnace. The container shall be cast or fabricated from nickel-chromium alloy, and shall be capable of supporting maximum furnace loads at maximum operating temperature. The bottom of the container shall be either a cast or fabricated grid. All work containers shall include a wire mesh screen or perforated metal plate on top of the bottom grid which has sufficient open area to allow adequate circulation through the work area. Work baskets shall be designed so that maximum loads can be supported form the bottom grid or from a spider type fixture supported from the top of the basket. When specified (see 6.2.1), the containers shall consist of individual trays capable of being stacked one on top of the other. Trays shall be provided with removable bottoms and means for locking together for handling the stack as a single container. Tray dimensions shall be as specified (see 6.2.1). All work containers and lifting devices shall be equipped with lifting eyes, rings or trunnions which are capable of handling maximum loads. When required, trays which fit inside the work basket shall be provided. Tray depth shall be as specified (see 6.2.1).

3.7.5 Heating elements. The heating elements shall be designed and located to meet the heating requirements and temperature uniformity specified herein. Class 1 and class 2 furnaces, that are designed to operate without controlled atmosphere, shall be provided with heating elements with an alloy composition of a least 35 percent nickel and 20 percent chrome. All furnaces designed for controlled atmosphere operation, and all class 3 furnaces, shall be provided with heating elements containing an alloy composition of 80 percent nickel and

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20 percent chrome. Power density on the heating elements shall not exceed 30 watts per square inch for the class 1 and class 2 furnaces, or 20 watts per square inch for the class 3 furnace. The terminals of the heating elements shall be designed to have at least twice the current carrying capacity of the heating element. Solderless connections to power leads shall be furnished. Provisions shall be included to protect the elements from damage during inserting and removing containers, and to prevent loose parts from falling in the element area.

3.7.6 Recirculating system. The recirculating system shall be capable of producing and maintaining the temperature uniformity specified in 3.6.2 for each class furnace. The direction of air or atmosphere flow shall be from the top to the bottom of the work chamber. The fan drive motor shall be in accordance with 3.4.2. The system shall provide for automatically shutting off the power to the heating elements, if the fan stops for any reason.

3.7.7 Heat exchanger. When specified (see 6.2.1), the furnace shall be designed for and provided with an external water cooled heat exchanger and auxiliary blower system to expedite the cooling rate of the work load. When specified, the system shall be of the closed cooling system of either the refrigeration or cooling tower design as specified.

3.7.8 Thermocouples. Chromel-alumel type thermocouples conforming to ISA Standard MC96.1 shall be used as the temperature sensing device for each instrument. The thermocouple lead wires shall be at least 25 feet in length, and of the same material as the thermocouples; or of a material having the same thermo-electric characteristics at a thermocouple head temperature between 0 degrees and 250 degrees Fahrenheit.

3.7.9 Instruments. The instrument shall have output receptacles for temperature recorder and alarm systems. Instruments using thermocouples as the temperature sensing device shall be marked to indicate the particular type of thermocouple for which it is calibrated. Automatic reference junction compensation shall be provided in all instruments using thermocouples as the primary temperature sensing element.

3.7.9.1 Microprocessor. The furnace control system shall incorporate a microprocessor digital programmable/controller to provide maximum control flexibility within the limitations of the furnace's designed capability. The microprocessor shall provide for not less than 50 read/write programmable segments of random access memory and input into the system shall be by manual data input (MDI) through a keyboard and magnetic tape cassette. Additional or different program capability shall be provided when specified (see 6.2.1).

3.7.9.2 Temperature controller. Unless otherwise specified, the controller shall be of the time proportioning control system and shall operate a contactor through an intermediate relay device. When required, the controller shall be of the current proportioning type control, and shall operate either a variable reactance transformer or a saturable core reactor or silicon controlled rectifier, as specified (see 6.2.1). Either system furnished shall have the necessary adjustments to be capable of maintaining the set control

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temperature within ±2 degrees Fahrenheit and have an accuracy of 1/2 percent of set point.

3.7.9.2.1 Digital display. The indications of the control shall be direct reading in degrees Fahrenheit with a resolution of not more than 1 degrees Fahrenheit. The instrument shall provide simultaneous display of temperature set point and actual temperature with individual digits of the actual temperature not less than 0.56 inch high. The controller shall provide key touch selections of functions and the keyboard shall be membrane-covered to seal out dust and vapors.

3.7.9.3 Temperature recorder. Unless otherwise specified, the furnace shall be provided with a recorder of the strip chart self-balancing type, with calibrated chart width of not less than 6 inches and a chart length of not less than 80 feet to the roll with a variable speed drive. When specified (see 6.2.1), the chart width shall be not less than 9 inches and the chart length not less than 120 feet to the roll.

3.7.9.3.1 Process reporter. When specified (see 6.2.1), the furnace shall be provided with a digital process reporter in lieu of a strip chart recorder.

3.7.9.4 Alarms control. The controller shall provide for selection high low alarm control through the temperature control range of the furnace. The controller shall also provide thermocouple failure alarm or open circuit alarm conditions.

3.7.9.5 Instrument panel. Floor mounted instrument panel(s) conforming to NEMA ICS 1 shall be provided for mounting all instruments and accessory controls. A separate enclosure shall be provided for mounting the power contactor individually and separate from the instrument panel. All ventilated openings shall be designed to prevent entrance of any deleterious substance. Signal lights shall be provided to indicate the furnace cycle performance. The panel shall provide flush or semi-flush mounting of all instruments.

3.8 Flowmeter. When specified (see 6.2.1), a flowmeter shall be provided to regulate atmosphere flow.

3.9 Dewpoint analyzer. When specified (see 6.2.1), an automatic dewpoint analyzer shall be provided to analyze the atmosphere dewpoint during operation of the furnace.

3.10 Work baskets and fixtures. When specified (see 6.2.1), fixtures and extra work containers shall be furnished. Number required, design, and dimensions shall be as specified.

3.11 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.12 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

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3.13 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate, marked in accordance, shall be securely attached to each furnace. The nameplate shall contain the information listed below. If the furnace is a special model, the model designation shall include the model of the basic standard furnace and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and ratings
Inside work chamber dimensions
Operating temperature range (degrees F)
Type of atmosphere
Date of manufacture
Contract Number or Order Number
Federal Stock Number or Plant Equipment Code

3.14 Technical data. When technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, machine alignment and accuracy test results) is required it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.15 Workmanship. Workmanship of the furnace shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.4).

b. Acceptance test (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all quality conformance inspections, tests, and examinations (see 4.4) shall be performed in the manufacturer's designated indoor test area under normal environmental conditions and all acceptance test requirements (see 4.5) shall be performed only after the grinder is installed at its final location.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination or all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.1, and 4.7.4, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The furnace and equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment, safety conformance, and workmanship to determine compliance with the requirements of this specification.

4.7 Tests. All tests shall be conducted at an ambient temperature of 77 degrees Fahrenheit ±18 degrees Fahrenheit. Test instruments and sensing elements used to perform the test specified for the furnace and instruments should have been checked against a standard potentiometer type instrument, and calibrated thermocouple of known accuracy, within the previous 6 months to read accurately within ±2 degrees Fahrenheit.

4.7.1 Instrument test. Indicating and recording instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where normally the thermocouples would be connected. These tests shall be made in accordance with, and results checked against, standards established in ANSI C96.1.

4.7.2 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance, and faulty grounds, in accordance with the requirements of NEMA ICS 1.

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4.7.3 Operational test. The furnace shall be prepared for operation, and operated in accordance with manufacturer's handbook of instruments. The furnace shall be operated empty at 600 degrees Fahrenheit (class 1 and 2) and at 1200 degrees Fahrenheit (class 3) to check operation of control system and fan rotation direction. The fan interlock, and all safety devices, shall be tested by simulating failures that normally would cause the device or system to operate. Each system shall perform its intended function. The cover lift mechanism shall be checked, and the effort required to lift, swing, or close the cover shall not exceed a force of 40 pounds. Upon completion of the above test, the empty furnace shall be stabilized at the recovery temperatures listed in 3.5.1 for a minimum of 3 hours. After equalibrium has been established, the furnace shall be loaded with a charge of iron or steel equivalent to at least the recovery rate specified in table II. At the end of 1 hour, the chamber temperature shall have recovered and stabilized at the initial temperature within ±2 degrees Fahrenheit. Furnaces equipped for programmed cycles or heat exchanger shall be operated through a complete cycle. The presence of leaks, or any damage to the furnace or control equipment, shall be cause for rejection.

4.7.4 Performance tests. The furnace shall be loaded with a charge of iron or steel equivalent to at least 50 percent of maximum load specified for the size furnace, and a chamber temperature survey shall be performed. A minimum of nine test locations shall be used, with eight thermocouples located symmetrically through the load so as to scan the work space. One thermocouple shall be located within 6 inches of the control couple. For furnaces with a chamber volume of less than 10 cubic feet, the test may be performed with a minimum of three thermocouples, with one each located in the top, middle and bottom of the chamber. The survey shall be performed at the minimum and at the maximum operating temperature of the furnace. Readings should be continuously recorded or taken frequently enough to determine when the temperature of the test zone approaches the bottom of the temperature range being surveyed. From the time until thermal equilibrium is reached, none of the temperature readings should exceed the maximum of the range being surveyed. After thermal equilibrium is reached, readings shall be taken at not greater than 5 minute intervals for at least 30 minutes, and the maximum temperature variation of all points under test. With the temperature stabilized at maximum operating temperature, the outside surface shell and cover temperature shall be checked at random locations, and shall not exceed the limits specified for the class furnace.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified, preservation, packing, and marking shall be accordance with MIL-F-3296. Level of preservation, and packing shall be specified (see 6.2.1).

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6. NOTES

6.1 Intended use. Furnaces covered by this specification are intended primarily for use in heat treat operations such as tempering, annealing, hardening, or normalizing metal parts and materials.

6.2 Ordering Data.

6.2.1 Procurement requirements. Purchases should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number and date of this specification.

b. Class, required (see 1.2).

c. First article approval, if required (see 3.1).

d. Specify, if fan may be located in cover (see 3.2).

e. Type of control atmosphere (endothermic, steam, ammonia gas, etc.), when required (see 3.2.1).

f. If furnace is required to be configured in a specific measurement system (US or SI), stated required system (see 3.2.2).

g. Painting, if different (see 3.3.5).

h. Power supply, voltage and phase (see 3.4.1).

i. Size required (diameter and depth in inches) (see 3.5).

j. Recovery rate and maximum load, if size is different from size listed in table II (see 3.6.1).

k. Air activated lifting device if required (see 3.7.2).

l. Baffle or work container, specify (see 3.7.4).

m. Tray type container, when required specify tray dimensions (see 3.7.4.2).

n. Work trays to fit inside a solid side wall container, specify tray depth (see 3.7.4.2).

o. Heat exchanger, (specify type of closed system, if required) when required (see 3.7.7).

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p. Additional program control, when required, specify program requirements such as purge, heating and cooling cycles, heating and cooling rates, and atmosphere control requirements (see 3.7.9.1).

q. Type of temperature controller, if different (see 3.7.9.2).

r. Recorder chart width, if different (see 3.7.9.3).

s. Process reporter, in lieu of recorder, if required (see 3.7.9.3.1).

t. Flowmeter, if required (see 3.8).

u. Dewpoint analyzer, if required (see 3.9).

v. Extra work baskets, containers, or fixtures, when required (see 3.10).

w. Repair parts, when required (see 3.9).

x. Nameplate, if different (see 3.12).

y. Inspection conditions, if different (see 4.3).

z. Quality conformance inspection, if different (see 4.5).

aa. First article inspection, if different (see 4.4).

bb. Level of preservation and packing, if required to be in accordance with MIL-F-3296 (see 5.1).

6.2.2 Consideration of data requirements. The requirements for technical data should be considered when this specification is applied on a contract. The Data Item Descriptions (DIDs), DD Form 1664, for required data should be listed on a Contract Data Requirements List (DD Form 1423). The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements List, (AMSDL) should be researched for the applicable DD Form 1664 and to ensure that only current approved DIDs are cited on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting office should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Material for tests. The quantity of material, required for operational and performance tests, is generally not available at manufacturer's plants. Also, before testing, the refractory and insulation must be completely cured which increases the possibility of damage in movement and transit of the furnace. For these reasons, consideration should be given to performing first article testing after the furnace has been installed at the site of operation.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (key word) listing:

Controlled atmosphere
Forced convection
Furnace
Pit type
Temperature uniformity

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TABLE I. Sizes and capacities.

Size, furnace, working (dimensions in inches)					
Diameter	Depth	Diameter	Depth	Diameter	Depth
12	15	20	60	35	48
12	24	24	24	35	60
15	30	24	36	35	72
20	24	24	48	48	48
20	36	24	60	48	60
20	48	35	36	48	72
				48	108

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TABLE II. Recovery rate and maximum load (not less than).

Working Chamber (Dimensions in inches)		Class 1	Class 2	Class 3	
Diameter	Depth	Recovery rate, lb. steel/hr. to 1000° F	Recovery rate, lb. steel/hr. to 1100° F	Recovery rate, lb. steel/hr. to 1100° F	Maximum loads (pounds)
12	15	105	150	125	300
12	24	210	300	200	500
15	30	290	375	300	700
20	24	385	550	350	700
20	36	385	550	500	700
20	48	420	600	550	1600
20	60	490	700	600	1600
24	24	490	700	525	2000
24	36	560	800	700	3000
24	48	600	850	650	300
24	60	630	900	600	3000
35	36	630	900	625	4500
35	48	630	900	775	4500
35	60	700	1000	725	4500
35	72	700	1000	925	4500
48	48	700	1000	950	6500
48	72	875	1250	1000	7000
48	108	1000	1600	1300	8500

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Custodians:
Navy - SH
Air Force - 99

Review activities:
Air Force - 84

User activities:
Army - AV, SM
Navy - OS, AS

Preparing activity:
DLA-IP

(Project Number:
3424-0112)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-80133B	Furnaces, Heat Treating, Pit Type, Electric, Forced Convection

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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